

Source: ode_inputs_cnn/final_ode_mu_inputs/README.md

Final ODE Mu Inputs

Purpose

This folder collects the final `mu(t)` candidate inputs from the image-factor sprint.

The scope is limited to signal construction and handoff. ODE solver integration, portfolio weight optimization, and fund-level policy constraints are not included here.

Folder Location

Repository root:

```
/home/jonghan/findalpha/delta/sprint2
```

Final handoff folder:

```
ode_inputs_cnn/final_ode_mu_inputs/
```

Main recommended file:

```
ode_inputs_cnn/final_ode_mu_inputs/selected_mu_input.csv
```

Recommended Input

item	value
Recommended input	<code>mu_image_factor_rank</code>
Source model	<code>ensemble_4family+image_factor_pc1</code>
Signal type	<code>cross_sectional_centered_rank</code>
Chart variant	<code>ohlc_ma_volume</code>
Lookback / horizon	<code>60 / 20</code>
OOS rows per input	<code>20,160</code>
OOS dates	<code>2,880</code>
Assets	<code>7</code>

The recommended input is an expected-return ranking signal. It should be treated as a `mu(t)` candidate, not as a standalone trading rule.

Why This Input

The original no-image baseline is weak. Adding path representation improves the ranking signal.

signal / model	interpretation	rank corr	Sharpe	cumulative return	hit rate
logistic_cumulative_scale	ETF raw/cumulative-return baseline, no image, no CNN/LSTM	-0.0072	0.0764	-0.0512	0.5556
Equal-weight ETF buy-and-hold	no model, no signal, 7-asset daily equal weight over OOS period	n/a	0.2086	0.2186	0.5257
logistic_image_scale	image-style scaling but simple logistic model	0.0392	0.3850	0.6569	0.5903
ensemble_4family	rank-corr baseline before image-factor add-on	0.0674	0.5430	1.2395	0.6042
ensemble_4family+image_factor_pc1	recommended final image-factor ranking input	0.0798	0.3155	0.5517	0.5694

Interpretation:

- The no-image raw ETF baseline has almost no cross-sectional prediction ability.
- Image-style representation improves rank correlation even with a simple logistic model.
- The final image-factor ensemble improves ranking quality over ensemble_4family .
- The final image-factor input lowers simple top-k Sharpe, so it should be tested inside the ODE optimizer rather than treated as a direct trading strategy.

Candidate Inputs In This Folder

input_name	source_model_name	role
mu_image_factor_rank	ensemble_4family+image_factor_pc1	Recommended ranking-improved $\mu(t)$ input
mu_image_factor_balanced	ensemble_best+image_factor_pc1	Secondary image-factor input with smaller Sharpe trade-off
mu_rank_baseline	ensemble_4family	Rank-correlation baseline before image-factor add-on
mu_sharpe_baseline	ensemble_best	Sharpe-stable baseline from previous ensemble sprint

Performance Snapshot

input_name	source_model_name	rank corr	Sharpe	rank corr lift	bootstrap CI
mu_image_factor_rank	ensemble_4family+image_factor_pc1	0.0798	0.3155	0.0124	[0.00001, 0.02505]
mu_image_factor_balanced	ensemble_best+image_factor_pc1	0.0731	0.5109	0.0125	[0.00057, 0.02448]
mu_rank_baseline	ensemble_4family	0.0674	0.5430	0.0000	baseline
mu_sharpe_baseline	ensemble_best	0.0606	0.6425	0.0000	baseline

Files

file	use
selected_mu_input.csv	Recommended input only: mu_image_factor_rank . Start here for ODE integration.
final_mu_inputs_long.csv	Main multi-candidate handoff file. One row per date-asset-input .
final_mu_inputs_wide.csv	Convenience wide table with all candidate inputs on each date-asset row.
sigma_wide.csv	Daily Sigma(t) in wide covariance format, aligned to the selected mu dates.
sigma_long.csv	Daily Sigma(t) in long matrix-entry format: one row per date-asset_i-asset_j .
returns_for_ode.csv	Realized daily asset returns aligned to the selected mu/Sigma dates.
input_performance_summary.csv	Rank-corr, Sharpe, cumulative return, hit rate, turnover, bootstrap CI.
handoff_summary_ko.md	Korean handoff summary for the team.
manifest.json	Machine-readable metadata.
sigma_manifest.json	Machine-readable Sigma metadata.

Sigma(t) Inputs

Sigma is already available in this folder.

file	shape	meaning
sigma_wide.csv	2,880 x 29	One row per date. Contains 7 variances and 21 off-diagonal covariances.
sigma_long.csv	141,120 x 5	One row per matrix entry. Contains full 7 x 7 covariance matrix per date.
returns_for_ode.csv	2,880 x 8	Daily realized returns aligned to mu(t) and Sigma(t) .

Sigma details:

- Source: ode_inputs_cnn/ensemble_best/ode_bundle.csv
- Scale: daily log-return covariance
- Rolling window: 60 trading days
- Date range: 2014-09-24 to 2026-03-09
- Assets: 7
- Missing values: none after alignment

The Sigma estimates are not model-specific to the image factor. They are rolling historical covariance estimates aligned to the final ODE mu input dates.

Related Evidence Files

purpose	path
Phase 1 image-factor significance report	ode_inputs_cnn/image_factor_extension/image_factor_report.md
Phase 1 image-factor significance CSV	ode_inputs_cnn/image_factor_extension/image_factor_significance.csv
Phase 2 ablation final report	ode_inputs_cnn/image_factor_phase12_final_report.md
Phase 2 ablation report	ode_inputs_cnn/image_factor_ablation/image_factor_ablation_report.md
Phase 2 ablation summary	ode_inputs_cnn/image_factor_ablation/ablation_summary.csv
Phase 2 ablation significance	ode_inputs_cnn/image_factor_ablation/ablation_significance.csv
Phase 2 ensemble add-on search	ode_inputs_cnn/image_factor_ablation/ablation_ensemble_search.csv
Original model comparison with no-image baseline	ode_inputs_cnn/comparison_with_baselines.csv

Column Guide

column	meaning
date	Signal date.
asset	ETF asset name.
input_name	ODE input candidate name.
source_model_name	Source signal/model used to create this input.
mu_signal	Primary ODE input. Date-wise centered cross-sectional rank.
mu_signal_type	Scaling method. Current value: cross_sectional_centered_rank .
mu_raw_score	Original ensemble/image-factor score before ODE handoff scaling.
mu_rank	Date-wise percentile rank of mu_raw_score .
mu_centered_rank	mu_rank demeaned within each date and input.
mu_zscore	Date-wise z-score of mu_raw_score .
future_return	Realized 20-day return. Evaluation only. Do not use this as an ODE input.
is_recommended	Whether the row belongs to the recommended input.
lookback	Historical lookback window used to create the signal.
horizon	Prediction horizon.
chart_variant	Image rendering variant behind the image factor.

How To Use In ODE

Recommended first experiment:

1. Read selected_mu_input.csv .
2. Pivot by date and asset .
3. Use mu_signal as the expected-return ranking input $\mu(t)$.
4. Read sigma_wide.csv or sigma_long.csv for $\Sigma(t)$.
5. Compare against mu_sharpe_baseline and mu_rank_baseline from final_mu_inputs_long.csv .

Minimal pandas sketch:

```
import pandas as pd

mu = pd.read_csv("ode_inputs_cnn/final_ode_mu_inputs/selected_mu_input.csv", parse_dates=["date"])
mu_wide = mu.pivot(index="date", columns="asset", values="mu_signal")

sigma = pd.read_csv("ode_inputs_cnn/final_ode_mu_inputs/sigma_wide.csv", parse_dates=["date"])
returns = pd.read_csv("ode_inputs_cnn/final_ode_mu_inputs/returns_for_ode.csv", parse_dates=["date"])
```

Interpretation Caveat

`mu_image_factor_rank` improves cross-sectional ranking quality versus `ensemble_4family`, but it does not improve simple top-k Sharpe. The correct interpretation is:

> The image factor adds useful expected-return ranking information. Whether that information improves final portfolio performance should be tested inside the ODE optimizer with risk control.